Exploiting Memory-Based Cross-Image Contexts for Salient Object Detection in Optical Remote Sensing Images

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***Abstract*— Current state-of-the-art methods for salient object detection in optical remote sensing images (RSI-SOD) primarily relies on individual image context to detect salient objects. However, the potential of cross-image contexts remains largely unexplored in existing works, which can provide valuable auxiliary and complementary information for discriminating object representations in RSIs. In this article, we investigate the utilization of cross-image contextual information for RSI-SOD. We propose a novel memory-based context propagation network (MCP-Net) to harness dataset-level contextual information. MCP-Net incorporates a cross-image dual memory (CDM) module to store dataset-level information and utilize it to generate contextual information for the current image. CDM effectively captures intra-scene variations by leveraging both foreground and background memory banks, resulting in improved object representations. Additionally, we enhance the representations by leveraging scale-aware context information within individual images. To preserve RSI details before memory modules, we introduce a shared attention-guided fusion (SAF) module to align the adjacent network-level features. Extensive evaluation results demonstrate the superior performance of our proposed method compared to state-of-the-art methods on three public benchmarks. These results affirm that the inclusion of cross-image contexts can significantly benefit salient object detection (SOD) in remote sensing images (RSIs).**

***Index Terms*— Optical remote sensing images (RSIs), salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) is a crucial step in several vision tasks, including image and video compression [1], image retargeting [2], image quality assessment [3], and object recognition [4]. It aims to identify the most attention-grabbing objects in an image. Over the past decade, SOD in natural scene images (NSI-SOD) has received significant attention, and numerous methods have been proposed [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15]. These methods have contributed to the advancement of NSI-SOD techniques.

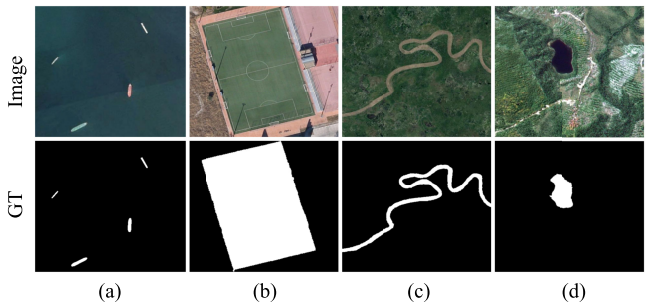


Fig. 1. Some typical optical RSIs and the corresponding salient object labels. (a) Multiple tiny objects. (b) Extremely large object. (c) Complex topology. (d) Cluttered background.

SOD in remote sensing images (RSI-SOD) poses specific challenges due to factors such as varying scales, complex topologies, cluttered backgrounds, and unstable imaging conditions. Fig. 1 showcases representative samples of remote sensing images (RSIs) that illustrate the challenges encountered in this domain. Such challenges have motivated researchers to develop specialized methods for RSI-SOD. Early methods focused on designing handcrafted features based on saliency priors, such as contrast [16] and sparsity representations [17], to detect salient objects. In recent years, the application of convolutional neural networks (CNNs) has significantly improved the performance of RSI-SOD. One pioneering method, LVNet [18], proposed by Li et al., introduced a nested structure that fuses multiple resolution inputs, enabling the detection of objects of various sizes in RSIs. Additionally, they constructed the first large dataset for RSI-SOD, which has facilitated subsequent research in the field. To enhance the representation for accurate salient object detection (SOD) in RSIs, the following works have focused on different aspects, including multiscale feature integration [18], [19], [20], edge preservation [21], [22], [23], and attention-based context aggregation [24], [25], [26].

Although various methods have been proposed to advance RSI-SOD, most of them focus on exploiting contextual information within individual images. However, cross-image

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contextual information is crucial to explore due to the intrinsic characteristics of salient objects in RSIs. Even salient objects of the same type exhibit significant variations in sizes, topologies, etc., across different RSIs. Additionally, salient objects tend to blend more with background regions in RSIs. Relying solely on the context of an individual image would result in inadequate and indiscriminate feature representations. Furthermore, CNN-based models for SOD predict pixel-level representations from the perspective of the entire dataset rather than a single input image. Consequently, the feature learning process is biased toward frequently occurring objects rather than rarely occurring ones. This limitation underscores the necessity of incorporating semantic information from corresponding categories in other images to achieve accurate and comprehensive SOD.

Memory networks have proven to be effective in storing and retrieving historical information for various tasks, including semantic segmentation [27], [28] and anomaly detection [29]. Jin et al. [27] explored the contextual information beyond the input image by aggregating the dataset-level representations of different pixel categories, resulting in improved pixel-level representations for semantic segmentation. Park et al. [29] proposed a memory module for anomaly detection, which records diverse prototypical patterns of normal data. However, applying existing memory modules to RSI-SOD is currently impractical due to several challenges. First, SOD is inherently class-agnostic, and existing datasets lack class annotations, making it infeasible to apply memory modules designed for class-specific tasks. Second, both foreground and background representations play vital roles in enhancing pixel-level representations for salient objects in remote sensing scenarios, further complicating the application of existing memory modules. Thus, novel approaches are required to develop memory networks specifically tailored for RSI-SOD, addressing the unique characteristics and challenges of this task.

In this study, we focus on harnessing the potential of cross-image contextual information for RSI-SOD. Fig. 2 illustrates the schematic difference between our method and previous approaches. Instead of relying solely on individual images, as done in previous methods, our approach aims to leverage memory mechanism for exploiting cross-image information. To this end, we propose a novel memory-based context propagation network (MCP-Net) for RSI-SOD. The core of MCP-Net is the cross-image dual memory (CDM) module, which explicitly stores and retrieves dataset-level information. It reduces learning bias toward frequently occurring objects in current CNN-based methods. The CDM incorporates distinct memory banks for foreground and background, dedicated to capturing scene-specific representations. This innovation not only expands the capabilities of existing memory networks but also ensures adaptability for the RSI-SOD task. In parallel, the intra-image scale-adaptive memory (ISM) propagation module is designed to fortify scale-aware context information within each image, complementing the cross-image context in CDM. Additionally, a shared attention fusion (SAF) module is introduced to align adjacent network-level features, which preserves intricate details and facilitates effective collaboration with memory modules. We extensively evaluate our proposed method on three public benchmarks, and the results verify the effectiveness of our methodology in detecting salient objects in diverse remote sensing scenarios.

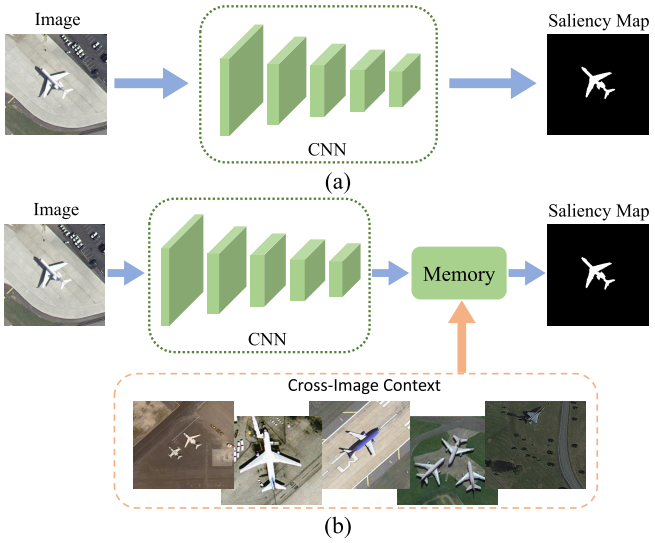


Fig. 2. Overall difference between (a) existing single-image-based methods and (b) our proposed cross-image-based method.

Our work makes the following distinct contributions.

1) We explore the utilization of cross-image contexts to generate discriminative pixel-level representations for RSI-SOD. By employing memory modules to store and retrieve information across images, our study sheds light on the significance of explicitly incorporating multi-image information for RSI-SOD.
2) We propose a novel framework, called MCP-Net. MCP-Net fully leverages both cross-image and intra-image contextual information to enhance salient object representations in RSIs.
3) We propose a CDM module within MCP-Net, to capture scene-specific foreground and background representations across the entire dataset. It provides a comprehensive understanding of the RSI scenes and the prominent objects inside.
4) We conduct extensive experimental evaluations on three public RSI-SOD benchmarks, demonstrating the superiority of our method over 19 state-of-the-art methods.

II. Related Works

A. SOD in Natural Scene Images

1) Traditional Methods: NSI-SOD has been extensively investigated in recent decades. Itti et al. [30] proposed a pioneering model for saliency detection in NSIs, which leverages center-surround contrast and multiscale image features inspired by human visual perception characteristics. Hou and Zhang [31] introduced a fast method for constructing saliency maps by extracting the spectral residual of an image in the frequency domain. Achanta et al. [5] proposed a method to extract salient object regions with well-defined boundaries, which are valuable for real-world vision applications. By focusing on regions with clear boundaries, their approach improves the interpretability and usability of SOD

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results. Subsequent to these pioneering works, the following handcrafted methods explore various perspectives to exploit underlying saliency information in NSIs. These perspectives include statistics priors [32], [33], color contrast [7], feature integration [34], and topology structure [10]. These methods utilize domain-specific insights and image properties to enhance the accuracy and robustness of SOD.

2) Deep Learning Methods: CNN-based methods have made significant advancements and demonstrated clear advantages over handcrafted methods in SOD. The majority of CNN-based methods concentrated on effective multilevel feature fusion [11], [12], [35], [36] for improved detection robustness. For instance, Hou et al. [11] proposed to construct densely connected structures between different network levels, achieving complementary fusion between high-level and low-level features. Zhao et al. [12] introduced a simple gated network to control interference between the encoder and decoder in a U-shaped network. Besides, some works explored edge guidance as an effective clue for refining salient object regions. Zhao et al. [14] utilized features learned from edge detection to explicitly enhance multilevel representations for SOD. Wu et al. [15] simultaneously refined the feature representations of salient objects and object edges in a collaborative way. Additionally, exploiting high-level features, e.g., extremely downsampled features [37], pyramid pooling features [13], etc., has been shown effective for locating salient objects. Recently, researchers have deployed Transformer [38] to NSI-SOD, enabling models to capture long-range dependencies effectively, thereby improving saliency detection [39], [40]. The effectiveness of these Transformer-based methods indicates the importance of extensive fields of view for SOD, inspiring our exploration that extends beyond the confines of a single image.

These approaches have significantly advanced NSI-SOD. However, as suggested in previous studies [20], [41], directly applying NSI-SOD methods to RSI-SOD is ineffective due to the unique characteristics of RSIs. Therefore, it is imperative to develop specialized RSI-SOD methods capable of achieving satisfactory results across diverse remote sensing scenarios.

B. SOD in Remote Sensing Images

1) Traditional Methods: RSI-SOD have been less explored compared to NSI-SOD. However, in recent years, there has been a growing interest in exploring SOD methods specifically tailored for RSIs. Traditional methods have primarily focused on designing handcrafted features to obtain discriminative representations for RSIs. Zhao et al. [17] proposed a sparsity-guided method that utilizes global and background cues for saliency integration. Jing et al. [42] proposed to calculate saliency maps in a top-down strategy by incorporating contour shape cues. Zhang et al. [43] exploited the intrinsic relationship between different cues, e.g., color, intensity, texture, etc., and proposed a self-adaptively multiple feature fusion model for saliency detection.

2) Deep Learning Methods: CNN-based methods have demonstrated superior performance to traditional handcrafted techniques in various aerial imagery applications, showcasing their robustness and versatility [44], [45], [46]. In recent years, researchers have developed CNN-based methods that leverage the power of CNNs for RSI-SOD. Li et al. [18] proposed LV-Net, the first CNN-based method for SOD in RSIs. LV-Net gathers multiscale representations from multiresolution inputs through a nested network architecture. Subsequent works have concentrated on extracting discriminative representations from individual images, mainly focusing on three aspects. We provide a brief overview of each.

1) *Multiscale integration:* Drawing on methods from NSI-SOD, numerous strategies for multiscale integration have been explored, such as adjacent level fusion [20], parallel down-up fusion [19], and semantic multiscale refinement [47]. Integrating multiscale features is crucial for handling the scale variability issue in RSIs, and has proven effective for SOD and land cover analysis, from large water bodies to small agricultural plots.
2) *Attention mechanism:* The attention mechanism is capable of modeling long-range dependencies by focusing on the most relevant features in RSIs, thereby reducing background noise and enhancing feature relevance. Zhang et al. [24] incorporated long-range attention mechanisms and propagated them from shallow to deep layers in a cascaded manner. Cong et al. [25] proposed a relational reasoning network with parallel multiscale attention. Li et al. [41] explored the complementarity of multiple contents in RSIs and exhaustively combined multiple attention cues, such as foreground, background, edge, and global attention, for SOD.
3) *Edge guidance:* Notably, the utilization of object edge cues has proven to be extremely useful in RSI-SOD [21], [22], [23], [41]. For instance, Zhou et al. [22] addressed the coarse boundary issue by conducting multiscale feature integration under the explicit and implicit guidance of salient edge cues. In [21], they proposed to utilize edge information to sharpen the position cues, constructing a shared encoder and two parallel decoders consisting of an edge extraction branch and a feature fusion branch. Preserving clear object boundaries can facilitate the usage of SOD in remote sensing applications such as change detection and infrastructure analysis.

The aforementioned RSI-SOD methods primarily focus on utilizing within-image context to identify salient objects. However, they often struggle due to the complex blending of salient objects with background clutter and the diverse scales and shapes of objects in RSIs. Our method addresses these challenges by harnessing cross-image contextual information across the entire dataset, thereby improving the detection robustness of salient objects in RSI scenarios.

C. Memory Module

Memory networks [27], [28], [29], [48], [49] have demonstrated effectiveness in various computer vision tasks, including semantic segmentation, video object segmentation, and visual tracking. Space-time memory (STM) networks [48] were proposed to address video object segmentation by leveraging spatial–temporal attention on every pixel in multiple

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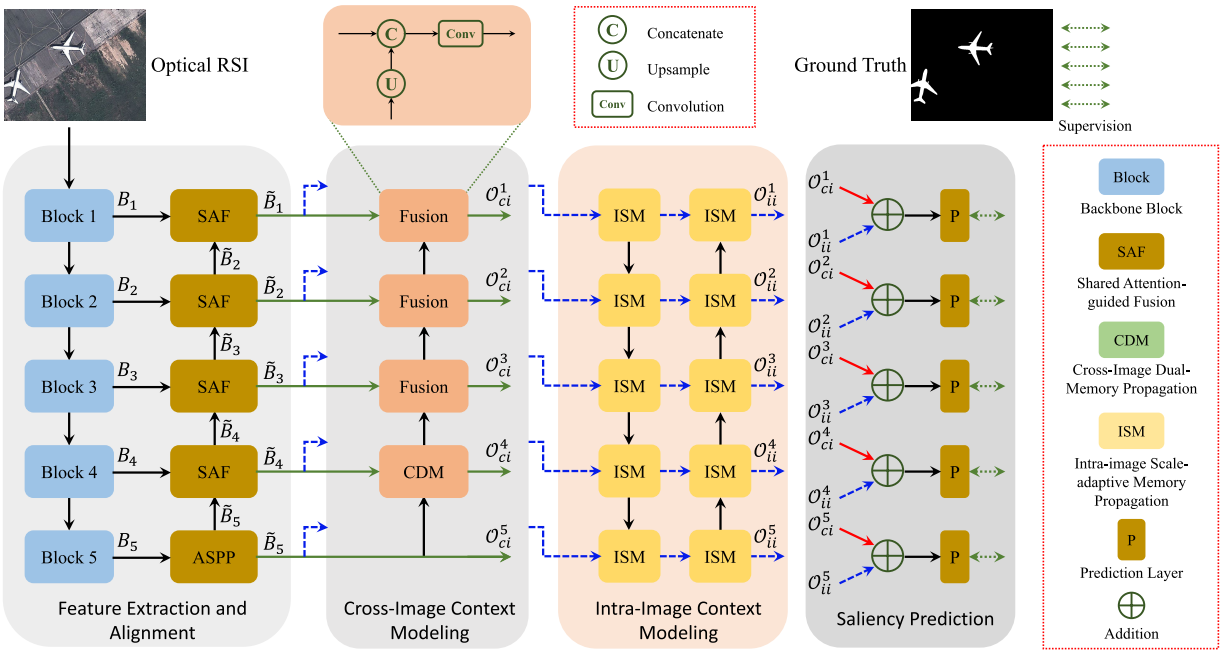


Fig. 3. Illustration of the overall architecture of the MCP-Net. MCP-Net comprises four main stages: feature extraction and alignment, cross-image context modeling, intra-image context modeling, and saliency prediction.

frames of a video, enhancing the feature representation of the current frame. Fan and Zhang [28] introduced a memory-based module to extract cross-image information for weakly supervised semantic segmentation. They employed a memory bank to store the class-aware features generated by image-level labels and used them to enhance the context of the current sample. MCIBI/MCIBI++ [27], [49] utilized dataset-level representations of various object categories for semantic segmentation. Park et al. [29] employed a memory module to explicitly address the diversity of normal patterns, which is leveraged for anomaly detection. However, these memory networks mainly focus on memorizing object-centric representations while lacking in describing background patterns. This limitation makes them unsuitable for addressing the highly confusable regions between salient objects and background regions in RSIs. In this work, we harness the power of memory networks and integrate both foreground and background memory banks to better separate salient objects in RSIs. To the best of our knowledge, we are the first to exploit cross-image contextual information for SOD tasks.

III. Methodology

The overall architecture of the proposed MCP-Net is depicted in Fig. 3. MCP-Net comprises four main stages: feature extraction and alignment, cross-image context modeling, intra-image context modeling, and saliency prediction. In the first stage, multiscale features are extracted from the input optical RSI, and aligned using a shared attention-guided fusion (SAF) module. The SAF effectively aligns information from adjacent levels to improve feature representation. The outputs from the first stage are concurrently fed into the second stage and third stages to capture complementary contextual information. The second stage utilizes cross-image contexts to provide abundant and auxiliary semantic features

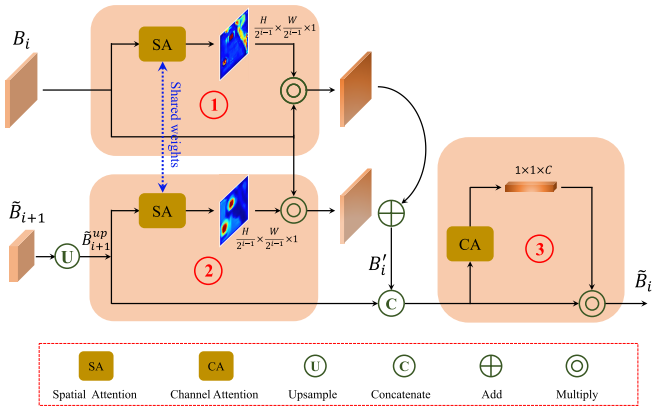


Fig. 4. Structure of the SAF module. ① Low-level guided detail preservation. ② High-level guided detail preservation. ③ Channel attention.

for the current image. The proposed CDM propagation module leverages information from RSIs across the entire dataset to enhance the understanding of the current image. In the third stage, intra-image contexts are enhanced through the use of a scale-adaptive memory module, called ISM propagation module. In our implementation, we stack two ISMs in each level to construct both bottom-up and top-down pathways, facilitating comprehensive scale information propagation across different network levels. In the last stage, the network integrates the feature representations derived from the cross-image context modeling and intra-image context modeling to predict the saliency maps. The network outputs five saliency maps appended at each network level, which are optimized using the ground-truth (GT) saliency map. In Sections III-A–III-D, we provide a detailed description of each of these four stages and elaborate on the newly designed modules that contribute to the effectiveness of MCP-Net.

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A. Feature Extraction and Alignment

We adopt VGG-16 [50] as the feature extraction backbone in our architecture for its straightforward design and multiscale representation capabilities. We keep the top five convolutional blocks of VGG-16 for feature extraction and discard the last pooling layer as well all fully connected layers. Given an input optical RSI $\mathcal{I} \in \mathbb{R}^{H\times W\times 3}$, we extract multiscale features from the five convolution stages of the backbone. These features are denoted as $\{B_i\}_{i=1}^{5}$, where each stage's spatial resolution is $(H/2^{i-1}) \times (W/2^{i-1})$. To capture high-level semantic information, we incorporate an atrous spatial pyramid pooling (ASPP) [51] after the fifth stage to further dig high-level semantic information. Next, following U-Net [52] and the feature pyramid network [53], we fuse multilevel features in a top-down manner, aligning high-level semantics with low-level details successively. However, conventional fusion operations, such as addition and concatenation, fail to effectively distinguish discriminative low-level details during the top-down fusion process, resulting in detail interference and redundancy. To address this issue, we introduce a novel SAF module and present its structure in Fig. 4. The SAF module is designed to preserve crucial detailed information by leveraging the guidance provided by both current and higher-level features. Specifically, a shared spatial attention layer is incorporated within SAF to ensure consistency between the predictions at the current level and higher-level features. This consistency is crucial as informative details should align with adjacent-level representations.

Let us denote the output of the SAF module as $\tilde{B}_i$. At each level, the SAF module takes two inputs: B_i, derived from the backbone network, and $\tilde{B}_{i+1}$, the output of the upper-level SAF module. The preservation of low-level details can be expressed as follows:

$$B_i' = \mathrm{SA}\left(\tilde{B}_{i+1}^{\mathrm{up}}\right) \odot B_i + \mathrm{SA}(B_i) \odot B_i \tag{1}$$

where SA represents the shared spatial attention generation function, implemented using a 3×3 convolution with a sigmoid activation. $\odot$ in (1) denotes location-wise multiplication. $\tilde{B}_{i+1}^{\mathrm{up}}$ denotes $2\times$ upsampled $\tilde{B}_{i+1}$. Next, the preserved low-level detail representation B_i' is concatenated with the higher-level feature $\tilde{B}_{i+1}^{\mathrm{up}}$, which is further enhanced using a channel attention mechanism

$$\hat{B}_i = \mathrm{CA}\left(\tilde{B}_{i+1}^{\mathrm{up}} \| B_i'\right) \odot \left(\tilde{B}_{i+1}^{\mathrm{up}} \| B_i'\right) \tag{2}$$

where CA represents the channel attention generation function, which is implemented using global average pooling, a 1×1 convolution, and a sigmoid activation applied sequentially. $\|$ denotes concatenation operation, and $\odot$ in (2) denotes channel-wise multiplication. To obtain the final output $\tilde{B}_i$ of the SAF, a residual connection is established by adding the concatenation of B_i and $\tilde{B}_{i+1}^{\mathrm{up}}$ to $\hat{B}_i$ (which is omitted in the figure). These operations collectively enable the SAF to preserve discriminant low-level details guided by both the current and higher-level features, resulting in an enhanced representation $\tilde{B}_i$ for subsequent stages of the MCP-Net architecture.

B. Cross-Image Context Modeling

Cross-image contexts provide valuable auxiliary information for enhancing intra-image context. In RSIs, salient objects often blend with the background, making individual image analysis insufficient for satisfactory results. In this section, we propose the CDM propagation module to harness cross-image contexts across the entire RSI dataset. The pipeline of CDM is illustrated in Fig. 5. Specifically, CDM comprises two memory banks: the foreground memory bank and the background memory bank. These memory banks are responsible for capturing scene-dependent foreground and background contexts, respectively, across different samples. It is important to note that in the cross-image context modeling stage of the MCP-Net, we apply CDM only at the fourth network level, as cross-image high-level semantic information is more beneficial for enhancing the current image. A simple fusion module, depicted in Fig. 3, is used to propagate the cross-image contexts from CDM to the lower network levels. In the following, we elaborate on the processes of the memory-based context aggregation and the memory content update within CDM.

1) Memory-Based Context Propagation: Memory-based context propagation involves querying relevant feature from the memory banks that hold historical information. These features are then propagated to enrich the representation of the current image. The memory banks are continuously updated during the training iterations, thereby providing a wealth of cross-image contextual information.

a) CDM propagation module: CDM module incorporates two memory banks (denoted as $\mathcal{M}_r$, where $r \in \{0, 1\}$ represents background/foreground) to store scene category-dependent representations across different samples. Each memory bank has a size of $M \times K \times C$, where M is the number of memory units, K is the total number of scene categories, and C is the dimension of the feature representation. For notation simplicity, we denote the input to the CDM as $\mathcal{F}$, which is the fusion of B_4 and $\tilde{B}_5$ (fusion omitted in Fig. 3). $\mathcal{F}$ is utilized to query and aggregate relevant representations from the memory banks. First, $\mathcal{F}$ is fed into the dual prediction head (DPH) to predict the foreground and background probabilities across different scene categories. The probability matrix, denoted as $Q_r \in \mathbb{R}^{(H\times W)\times K}$, is then used to aggregate relevant representations from the memory banks, expressed as

$$\mathcal{F}_{mr} = Q_r \otimes \mathcal{M}_r(m), \quad r \in \{0, 1\} \tag{3}$$

where $\otimes$ in (3) denotes matrix multiplication, and $\mathcal{M}_r(m) \in \mathbb{R}^{K\times C}$ represents the mth memory unit. The background- and foreground-aggregated representations, $\mathcal{F}_{m0}$ and $\mathcal{F}_{m1}$, respectively, are added element-wisely to generate $\mathcal{F}_m$. Next, the relations between $\mathcal{F}$ and $\mathcal{F}_m$ are leveraged to refine the feature representation of the current image. The relation-based refinement is calculated as

$$\mathcal{F}_{mo} = \mathrm{Softmax}[Q(\mathcal{F}), K(\mathcal{F}_m)]V(\mathcal{F}_m) \tag{4}$$

where Softmax denotes the softmax function, Q, K, and V represent the functions for generating query, key, and

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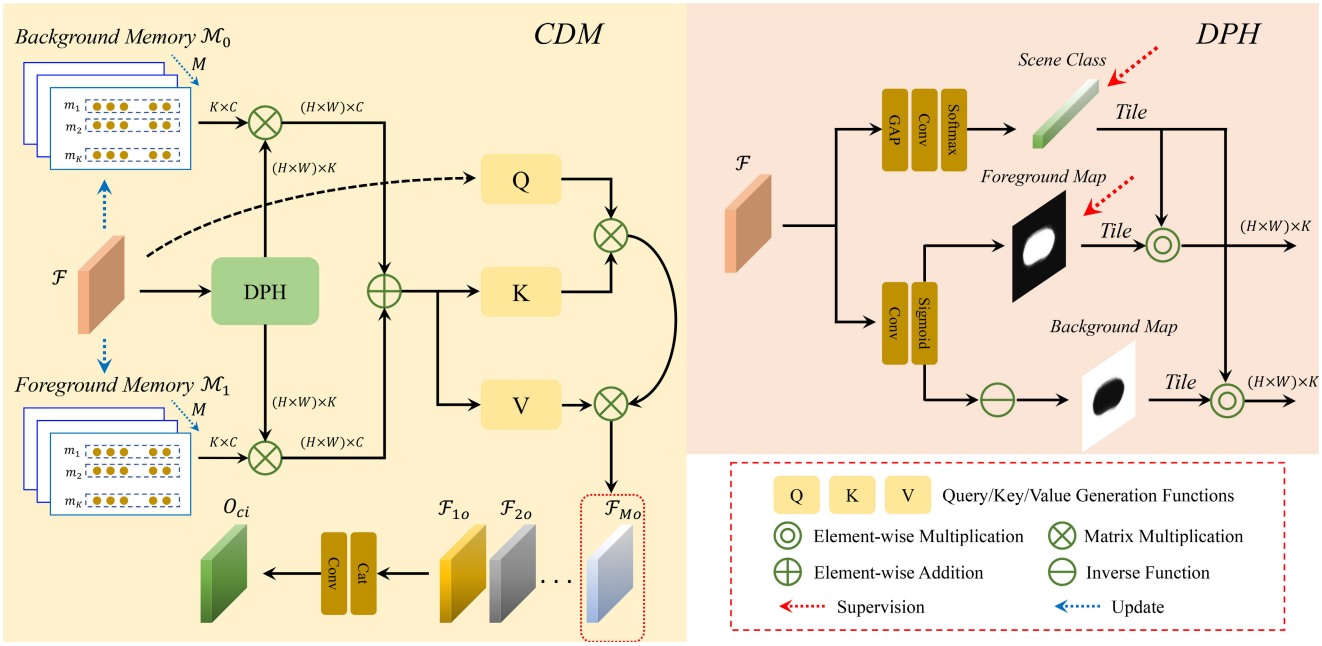


Fig. 5. Illustration of the CDM propagation module and the DPH.

value, respectively. These functions are implemented using a combination of convolution, batch normalization, and ReLU activation. Finally, the relation-refined features $\{\mathcal{F}_{mo}\}_{m=1}^{M}$ from all memory units are concatenated and linearly transformed to produce the output of the CDM.

b) Dual prediction head: The DPH predicts both scene distribution probability and foreground/background probabilities for the current-image representation. Initially, we obtain the whole-view-aware scene category probability $\mathrm{S}_c \in \mathbb{R}^{1\times1\times K}$ by applying global average pooling, followed by a 1×1 convolution and a softmax function. This yields a compact representation of the scene category probabilities. Then, a foreground probability map $\mathrm{S}_1 \in \mathbb{R}^{H\times W\times1}$ is generated by applying a 1×1 convolution and a sigmoid function. Simultaneously, S_1 is passed through an inverse function, which is implemented by subtracting each value in S_1 from 1. This operation generates the background probability map $\mathrm{S}_0 \in \mathbb{R}^{H\times W\times1}$. To predict the foreground/background probabilities under various scene categories S_c, we utilize the scene category probability S_c and the foreground/background probability maps S_1 and S_0. This is achieved through the following equation:

$$Q_r = \mathrm{Tile}_{\mathrm{spa}}(\mathrm{S}_c) \odot \mathrm{Tile}_{ch}(\mathrm{S}_r), \quad r \in \{0, 1\} \tag{5}$$

where $\odot$ denotes element-wise multiplication, $\mathrm{Tile}_{\mathrm{spa}}$ represents tiling the representation repeatedly to match the spatial resolution of $H \times W$, and Tile_{ch} represents tiling the representation repeatedly to match the channel dimension K. The resulting Q_r is resized to $\mathbb{R}^{(H\times W)\times K}$ and employed to query relevant representations from the memory banks. During training, the optimization of the scene probability representation S_c is supervised using scene-category annotations, while the optimization of the foreground and background predictions is supervised using annotated saliency maps. This joint supervision encourages the model to effectively utilize the contextual information stored in the memory banks, enhancing its ability to capture cross-image dependencies and improve overall prediction accuracy.

2) Memory Content Update: We utilize the acquired foreground/background representations, which depend on the scene categories, to update the memory banks. This process can be expressed as

$$\mathcal{F}_{rk} = \left\{ \mathcal{F}_{[i,:]}^{(H\times W)\times C} \,|\, \left(\mathrm{GT}_i^s = r\right) \wedge \left(\mathrm{GT}^c = k\right) \right\} \tag{6}$$

where GT^c denotes the scene category annotations, $k \in \{0, 1, \ldots, K-1\}$ is the scene category index, GT^s denotes the salient object annotations, and $r \in \{0, 1\}$ is the pixel-level foreground/background index. The input $\mathcal{F}$ is resized to $\mathbb{R}^{(H\times W)\times C}$ for the convenience of a feature collection. Consequently, $\mathcal{F}_{rk}$ stores the feature representations for foreground/background regions ($r \in \{0, 1\}$) under a certain scene category k. We randomly pick M images per scene category, and then use the mean feature representations of foreground and background regions to initialize the memory banks.

In RSIs, the contents within a specific scene category, such as object appearances and background patterns, still exhibit large variation across different samples. To capture these intra-scene variations, we employ M memory units. The memory representations for a given scene category across different units can be denoted as $\mathcal{M}_{rk} := \mathcal{M}_r(:, k, :) \in \mathbb{R}^{M\times C}$. To measure the similarity between each representation in $\mathcal{F}_{rk}$ and the corresponding representation in $\mathcal{M}_{rk}$, we calculate the cosine similarity, which can be expressed as

$$W_{rk}[i, j] = \frac{\mathcal{F}_{rk}[i] \cdot \mathcal{M}_{rk}[j]}{\left\|\mathcal{F}_{rk}[i]\right\|_2 \cdot \left\|\mathcal{M}_{rk}[j]\right\|_2} \tag{7}$$

where i and j represent the item indices in $\mathcal{F}_{rk}$ and $\mathcal{M}_{rk}$, respectively. $W_{rk} \in \mathbb{R}^{N_{rk}\times M}$ denotes the similarity

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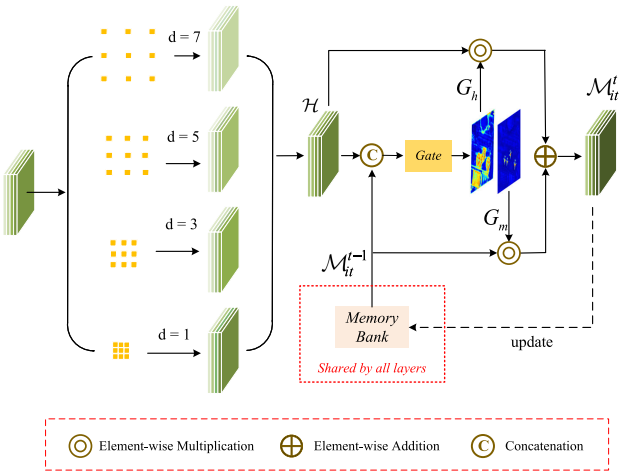


Fig. 6. Illustration of the ISM propagation module.

matrix, where N_{rk} is the number of pixels labeled as foreground/background r under the category k. The updated content is calculated as follows:

$$\tilde{\mathcal{M}}_{rk}[j] = \sum_{i=1}^{N_{rk}} \frac{1 - W_{rk}[i, j]}{\sum_{p=1}^{N_{rk}} (1 - W_{rk}[p, j])} \cdot \mathcal{F}_{rk}[i] \tag{8}$$

where $\tilde{\mathcal{M}}_{rk}[j]$ represents the updated content for the jth memory unit in $\mathcal{M}_{rk}$, where the contribution of each $\mathcal{F}_{rk}[i]$ is weighted by the complement of its cosine similarity with the corresponding memory representation.

Finally, during each training iteration, the corresponding item in the memory banks is updated using a moving average operation

$$\mathcal{M}_{rk}^{t}[j] = (1-m) \cdot \mathcal{M}_{rk}^{t-1}[j] + m \cdot \tilde{\mathcal{M}}_{rk}^{t}[j] \tag{9}$$

where m denotes the update momentum, and t denotes the iteration step. In our experiments, we empirically set m to 0.1 to strike a balance between preserving previous memory information and incorporating new information.

C. Intra-Image Context Modeling

Intra-image context modeling plays a crucial role in capturing fine-grained details and context-specific information within an individual RSI, complementing the cross-image contexts captured in the previous stage. To this end, we propose an ISM propagation module, which enables comprehensive scale-aware representations within each network layer, and propagate them across layers using an intra-image memory bank $\mathcal{M}_{it}$. The size of $\mathcal{M}_{it} \in \mathbb{R}^{h \times w \times c}$ is consistent with that of the feature map at each corresponding layer. The structure of the ISM module is illustrated in Fig. 6. It is designed to address the challenges posed by scale and shape variations typically observed in salient objects within RSIs. By performing multiscale learning within a single layer, the module is able to effectively capture and represent the diverse scales present in the image data. Moreover, the memory bank in ISM adopts a simple update strategy, allowing for efficient propagation of representations across different layers in a scalable manner.

Multiscale learning with an individual layer is achieved by an ASPP module [51], deriving a context-aware scale-adaptive feature $\mathcal{H}$. To update the memory representation, we employ a gate mechanism that selectively combines the current-layer multiscale representation with the memory representation $\mathcal{M}_{it}^{t-1}$ from the previous iteration. Herein, t denotes the iteration step. The memory update process involves calculating two gate attention maps, which determine the importance weights for the current layer and memory representations. The calculation can be expressed as

$$G_h,\ G_m = \text{Gate}(\mathcal{H} \| \mathcal{M}_{it}^{t-1}) \tag{10}$$

where $G_h and G_m$ are gate attention maps of size $H \times W \times 1$ and are generated by the Gate function, which is implemented using a 3×3 convolution followed by a sigmoid function. The memory representation $\mathcal{M}_{it}^{t-1}$ is resized to match the resolution of $\mathcal{H}$ before being fed into (10). Next, a location-wise combination is performed between $\mathcal{H}$ and $\mathcal{M}_{it}^{t-1}$ to update the memory representation as follows:

$$\mathcal{M}_{it}^{t} = G_h \odot \mathcal{H} + G_m \odot \mathcal{M}_{it}^{t-1} \tag{11}$$

where $\odot$ denotes location-wise multiplication. The memory representation is updated and transferred layer by layer, allowing scale-adaptive information to propagate between different levels of the architecture. The updated memory representation $\mathcal{M}_{it}^{t}$ also serves as the output of the ISM.

However, as the memory representation propagates in a single direction through multiple layers, it may become diluted. Moreover, psychological studies [54] suggest that both top-down and bottom-up processes are involved in human visual perception. Thus, we stack two ISM modules at each network level and construct bidirectional ISM connections, as depicted in Fig. 3, which strengthens the propagation of scale-adaptive context information for SOD.

D. Saliency Prediction and Optimization

We obtain five final saliency outputs at each network level. Denoting the outputs from the cross-image context stage as $\{\mathcal{O}_{ci}^{l}\}_{l=1}^{5}$ and the outputs from the intra-image context stage as $\{\mathcal{O}_{ii}^{l}\}_{l=1}^{5}$ (where l represents the layer index), the saliency predictions are derived as

$$\mathrm{S}^{l} = \sigma\left(\mathrm{UP}\left(\mathrm{Conv}_{1\times1}\left(\mathcal{O}_{ci}^{l} + \mathcal{O}_{ii}^{l}\right)\right)\right) \tag{12}$$

where σ represents the Sigmoid function, and UP denotes the bilinear upsampling layer that upsamples the maps to the original resolution of the input RSI.

To ensure better convergence, we supervise both the final outputs $\{\mathrm{S}^{l}\}_{l=1}^{5}$ and the intermediate representations, namely S_c, S_1. For the supervision of the final saliency predictions, we adopt two types of loss functions: binary cross-entropy (BCE) and intersection over union (IoU). The BCE loss is given by

$$\mathcal{L}_{\text{bce}}(\mathrm{S}, G) = -\sum_{i=1}^{H}\sum_{j=1}^{W}\big[G_{i,j}\log \mathrm{S}_{i,j} + (1-G_{i,j})\log(1-\mathrm{S}_{i,j})\big] \tag{13}$$

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where H and W denote the height and width of the RSI, respectively. S and G are the saliency prediction and the saliency GT, respectively. The IoU loss is calculated as follows:

$$\mathcal{L}_{\text{iou}}(\mathrm{S}, G) = 1 - \frac{\sum_{i=1}^{H}\sum_{j=1}^{W} \mathrm{S}_{i,j} G_{i,j}}{\sum_{i=1}^{H}\sum_{j=1}^{W} \left(\mathrm{S}_{i,j} + G_{i,j} - \mathrm{S}_{i,j} G_{i,j}\right)}. \quad (14)$$

For the supervision of the intermediate representations, we use cross-entropy (CE) loss to optimize the scene category prediction S_c and BCE loss to optimize the foreground prediction S_1. The CE loss is calculated as

$$\mathcal{L}_{ce}(\mathrm{S}, G) = -\sum_{k=1}^{K} G_k \log \mathrm{S}_k \quad (15)$$

where S and G are the scene category prediction and the corresponding GT respectively, and K denotes the number of scene categories.

The total loss of our proposed method can be defined as

$$\mathcal{L} = \mathcal{L}_{\text{bce}}(\mathrm{S}_1, G_s) + \mathcal{L}_{ce}(\mathrm{S}_c, G_c) \times \sum_{l=1}^{5} \left[\mathcal{L}_{\text{bce}}\left(\mathrm{S}^l, G_s\right) + \mathcal{L}_{\text{iou}}\left(\mathrm{S}^l, G_s\right) \right] \quad (16)$$

where l denotes the layer index, and G_s and G_c denote the saliency GT and scene category GT, respectively.

IV. Experimental Evaluation

A. Experiment Protocol

1) Datasets: We evaluate the performance of the proposed MCP-Net on three publicly available RSI-SOD benchmarks, including ORSSD [18], EORSSD [24], and ORSI-4199 [23]. ORSSD is the first dataset specifically constructed for the RSI-SOD task. It contains 600 images for training and 200 images for testing. EORSSD is an extended version of ORSSD, which contains 1200 images for training and 600 images for testing. ORSI-4199 is the largest RSI-SOD dataset available to data, comprising 2000 training images and 2199 testing images. All three datasets provide pixel-level annotations of salient objects for each image, enabling comprehensive evaluation of the performance of RSI-SOD methods.

2) Evaluation Metrics: We employ five commonly used evaluation metrics to assess the performance of our method and compare it with existing approaches, including F-measure [5], mean absolute error (MAE), S-measure [55], E-measure [56], and precision–recall (PR) curve. F-measure is calculates as the weighted average of precision and recall

$$F_\beta = \frac{\left(1+\beta^2\right)\text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}} \quad (17)$$

where β^2 is set to 0.3 to give greater weight to precision in SOD [11]. MAE measures the average absolute difference at all pixel locations

$$\text{MAE} = \frac{1}{H \times W} \sum_{h=1}^{H}\sum_{w=1}^{W} \|\mathrm{S}(h,w) - G(h,w)\| \quad (18)$$

where H and W denote the height and width of the input RSI, respectively. $\mathrm{S}(h,w)$ and $G(h,w)$ denote the saliency prediction and the corresponding GT at pixel location (h,w). S-measure evaluates the structural similarity between the predicted salient regions and the GTs. The S-measure is defined as

$$\mathrm{S}_m = \alpha * \mathrm{S}_o + (1-\alpha) * \mathrm{S}_r \quad (19)$$

where S_o is an object-aware similarity item and S_r is a region-aware similarity item. The former focuses on assessing the similarity of object-part regions, while the latter is designed to measure the entire object's similarity. α is set to 0.5 to assign equal contribution to both similarity items [55]. E-measure assesses the similarity between the predicted saliency maps and the GTs by considering local and image-level saliency values simultaneously. It is calculated using the alignment matrix ϕ and the enhanced alignment matrix $\theta(\phi)$ [56]

$$\mathrm{E}_\phi = \frac{1}{W \times H} \sum_{i=1}^{W}\sum_{j=1}^{H} \theta(\phi). \quad (20)$$

PR curve plots the precision rate against the recall rate by varying the threshold of the saliency map from 0 to 255. It provides a visual representation of the trade-off between precision and recall.

3) Implementation Details: The proposed MCP-Net is implemented using Pytorch and trained/tested on a server with an Intel Xeon Platinum 8375C CPU and an Nvidia RTX 3090 GPU. VGG-16 [50] is adopted as the feature extraction backbone for MCP-Net. The network parameters are optimized using the Adam optimizer [57]. During training, the batch size is set to 6, and the initial learning rate is set to 1×10^{-4}. After 30 epochs, the learning rate is divided by 10. To ensure consistent input resolutions, all input RSIs are resized to 256×256 before being fed into the MCP-Net model for both training and testing stages. Following common practices [20], [24], [58], our method is trained using augmented data, which includes flipping and rotation of the original training data. This augmentation strategy generates seven additional variants of the original training data. Additionally, we provide the scene category annotations for the three datasets, which are used for supervising the scene prediction layer as described in Section. III-B. The RSIs are classified into 15 scene categories, namely cars, ships, aircrafts, islands, roads, rivers, lakes, rural houses, urban landmarks, bridges, sports fields, industrial buildings, seas, monotonous scenes containing no salient objects, and others.

B. Comparison With State-of-the-Art Methods

We conduct a comprehensive comparison of our method with 19 state-of-the-art methods, including five CNN-based for NSI-SOD methods (i.e., DSS [11], RADF [35], EGNet [14], GateNet [12], PoolNet [13]), three traditional RSI-SOD methods (i.e., SMFF [43], VOS [59], CMC [42]), and 11 CNN-based RSI-SOD methods (i.e., LVNet [18], PDFNet [19], EMFINet [22], MJRBM [23], FSMINet [60], DAFNet [24], MCCNet [41], SARNet [61], SeaNet [62], ERPNet [21], ACCoNet [20]). The quantitative and qualitative comparisons of these methods are presented in the following.

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TABLE I

QUANTITATIVE EVALUATION. THE EVALUATION METRICS INCLUDE: MAE (M), STRUCTURE-MEASURE (S_m), MAXIMUM, MEAN, AND ADAPTIVE F-MEASURE ($F_\beta^{\max}$, F_β^{mean}, F_β^{ada}), MAXIMUM, MEAN, AND ADAPTIVE E-MEASURE ($E_\varepsilon^{\max}$, $E_\varepsilon^{\text{mean}}$, $E_\varepsilon^{\text{ada}}$). FOR MAE SMALLER IS BETTER, WHILE FOR THE OTHER METRICS LARGER IS BETTER. THE BEST THREE RESULTS ARE MARKED IN RED, GREEN, AND BLUE, RESPECTIVELY

Method	Param (M)↓	FLOPs (G)↓	Speed (FPS)↑	EORSSD								ORSSD							
				S_m↑	M↓	F_β^{max}↑	F_β^{mean}↑	F_β^{ada}↑	E_ε^{max}↑	E_ε^{mean}↑	E_ε^{ada}↑	S_m↑	M↓	F_β^{max}↑	F_β^{mean}↑	F_β^{ada}↑	E_ε^{max}↑	E_ε^{mean}↑	E_ε^{ada}↑
DSS	-	-	-	.7868	.0186	.6849	.5801	.4597	.9186	.7631	.6933	.8262	.0363	.7467	.6962	.6202	.8860	.8362	.8085
RADF	-	-	-	.8179	.0168	.7446	.6582	.4933	.9140	.8567	.7261	.8259	.0382	.7619	.6856	.5730	.9130	.8298	.7678
EGNet	108.1	313.9	43.5	.8601	.0110	.7880	.6967	.5379	.9570	.8775	.7566	.8721	.0216	.8332	.7500	.6452	.9731	.9013	.8226
GateNet	100.0	96.3	51.5	.9114	.0095	.8566	.8228	.7109	.9610	.9385	.8909	.9186	.0137	.8871	.8679	.8229	.9664	.9538	.9428
PoolNet	52.5	97.6	80.9	.8207	.0210	.7545	.6406	.4611	.9292	.8193	.6836	.8403	.0358	.7706	.6999	.6166	.9343	.8650	.8124
VOS	-	-	-	.5082	.2096	.2765	.2107	.1836	.5982	.4886	.4767	.5366	.2151	.3471	.2717	.2633	.6514	.5352	.5826
SMFF	-	-	-	.5401	.1434	.5176	.2992	.2083	.7744	.5197	.5014	.5312	.1854	.4417	.2684	.2496	.7402	.4920	.5676
CMC	-	-	-	.5798	.1057	.3268	.2692	.2007	.6803	.5894	.4890	.6033	.1267	.3913	.3454	.3108	.7064	.6417	.5996
LVNet	207.0	-	1.5	.8630	.0146	.7794	.7328	.6284	.9254	.8801	.8445	.8815	.0207	.8263	.7995	.7506	.9456	.9259	.9195
PDFNet	228.0	-	-	-	-	-	-	-	-	-	-	.9112	.0149	.8726	.8435	.7924	.9608	.9403	.9341
EMFINet	95.1	353.1	18.6	.9290	.0084	.8720	.8486	.7984	.9711	.9604	.9501	.9366	.0109	.9002	.8856	.8617	.9737	.9671	.9663
MJRBM	43.8	101.3	15.5	.9197	.0099	.8656	.8239	.7066	.9646	.9350	.8897	.9204	.0163	.8842	.8566	.8022	.9623	.9415	.9328
FSMINet	3.6	9.7	17.7	.9255	.0079	.8678	.8436	.8015	.9666	.9567	.9490	.9361	.0101	.9041	.8878	.8710	.9759	.9672	.9693
DAFNet	29.4	548.1	42.5	.9166	.0060	.8612	.7842	.6423	.9859	.9290	.8443	.9191	.0113	.8928	.8511	.7876	.9771	.9539	.9360
MCCNet	67.7	234.2	39.5	.9317	.0074	.8842	.8618	.8238	.9733	.9679	.9557	.9376	.0109	.9053	.8933	.8795	.9723	.9682	.9660
SARNet	25.9	137.3	43.4	.9182	.0115	.8695	.8373	.8030	.9597	.9514	.9431	.9048	.0206	.8733	.8489	.8346	.9511	.9436	.9405
SeaNet	2.7	2.6	107.6	.9150	.0083	.8560	.8435	.8184	.9666	.9602	.9559	.9197	.0124	.8786	.8599	.8483	.9643	.9595	.9587
ERPNet	56.5	227.4	73.7	.9172	.0098	.8629	.8275	.7349	.9615	.9310	.9125	.9182	.0160	.8832	.8624	.8278	.9601	.9406	.9465
ACCoNet	102.6	368.5	59.9	.9292	.0075	.8815	.8480	.7804	.9706	.9624	.9358	.9413	.0105	.9097	.8937	.8801	.9757	.9718	.9731
OURS	25.1	86.6	39.2	.9373	.0070	.8868	.8620	.8077	.9765	.9657	.9515	.9433	.0090	.9135	.8979	.8726	.9807	.9736	.9724

1) Quantitative Comparison: Table I presents the quantitative evaluations of S-measure (S_m), MAE, F-measure (F_β), and E-measure (E_ϕ) on ORSSD and EORSSD. The CNN-based methods designed for NSI-SOD perform better than the traditional handcrafted methods for RSI-SOD. However, they still fall behind the majority of CNN-based RSI-SOD methods, highlighting the necessity of specialized methods for RSI-SOD. In general, our proposed method achieves the best performance among all the CNN-based RSI-SOD methods. Specifically, our method ranks first in five metrics on ORSSD and in three metrics on EORSSD. While a few methods, such as DAFNet and MCCNet, exhibit better performance in certain metrics like MAE and maximum E-measure, they leverage object edge cues as extra information to refine salient object regions. On ORSSD dataset, our method improves the second-best method by 0.61%, 0.42%, 0.47%, and 10.9%, in terms of S_m, $F_\beta^{\max}$, F_β^{mean}, and MAE, respectively. On EORSSD, our method is better than the second-best method in terms of S_m (0.9373 against 0.9317), $F_\beta^{\max}$ (0.8868 against 0.8842), and F_β^{mean} (0.8620 against 0.8618). Table II provides the quantitative comparison on ORSI-4199 dataset, demonstrating that our method outperforms other competitive approaches. Given that ORSI-4199 is at least twice as large as the ORSSD and EORSSD datasets, the results confirm our method's scalability and efficacy in handling larger datasets. Moreover, ORSI-4199 dataset includes more challenging cases such as low contrast and incomplete salient objects, which demonstrates the robustness of our method in tackling more complex data.

Model complexity comparison is also provided in Table I. We evaluate model complexity and computational efficiency using three metrics: parameter count, floating-point operations (FLOPs), and frames per second (FPS). Our method demonstrates lower complexity than most of the compared methods in terms of parameter count and FLOPs, except for FSMINet and SeaNet, which are specifically designed for fast computation and lightweight applications. Notably, our method is significantly more lightweight and computationally efficient than well-performing methods like MCCNet. Our method consumes a 4170 M memory footprint and achieves a real-time inference speed of 39.2 FPS, which positions it in the mid-stream among all the compared methods.

TABLE II

QUANTITATIVE EVALUATION ON THE ORSI-4199 DATASET. THE BEST RESULTS ARE MARKED IN BOLD

Method	S_m↑	M↓	$F_\beta^{\max}$↑	$E_\varepsilon^{\max}$↑
ERPNet	.7867	.0663	.8683	.7594
MJRBM	.8593	.0374	.9311	.8493
SeaNet	.8634	.0345	.9357	.8562
MCCNet	.8686	.0338	.9346	.8592
ACCoNet	.8707	**.0314**	.9361	.8610
Ours	**.8736**	.0324	**.9402**	**.8667**

Fig. 7 illustrates the PR and F-measure curves on ORSSD and EORSSD. Our model's PR curves consistently exhibit superior performance compared to other methods, indicating advantages in predicting saliency across various thresholds. The F-measure curves further validate the superiority of our method. These curve evaluations demonstrate the robustness of our method in accurately estimating saliency in diverse situations.

2) Qualitative Comparison: In order to qualitatively compare our method with other state-of-the-art methods, we

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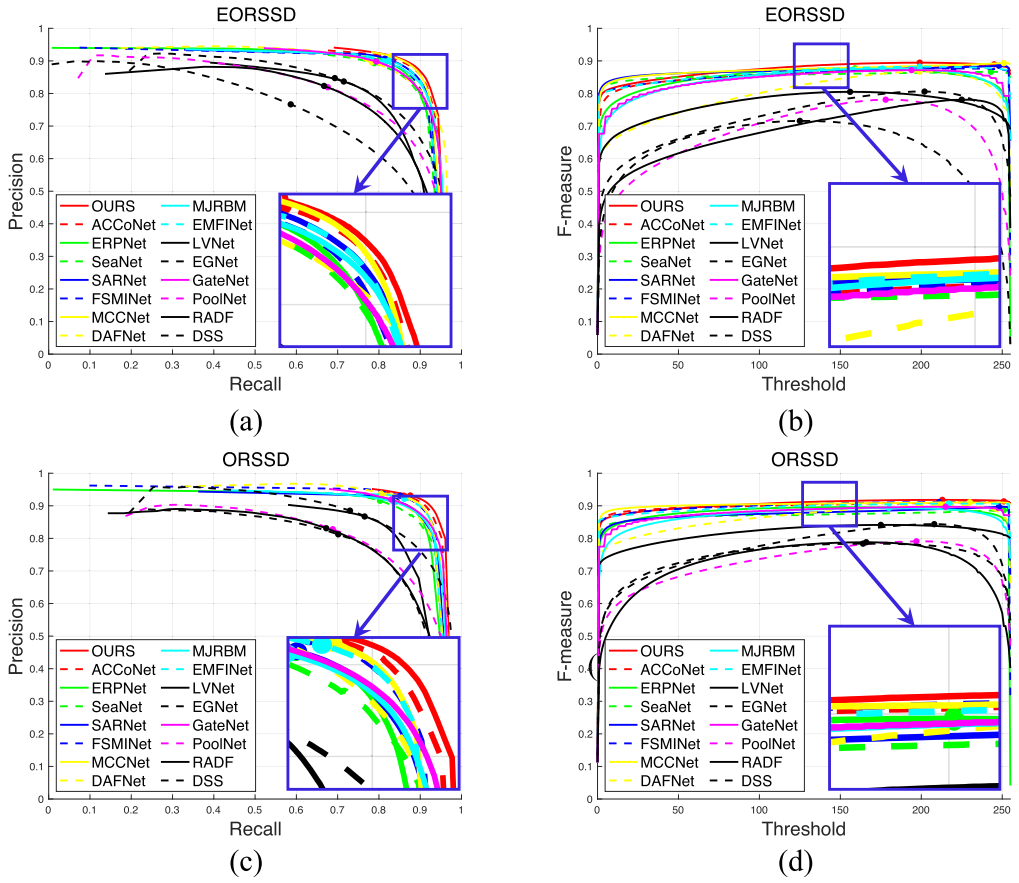


Fig. 7. Quantitative evaluation of different methods in terms of PR curve and F-measure curve. (a) PR curves on EORSSD. (b) PR curves on EORSSD. (c) F-measure curves on ORSSD. (d) F-measure curves on ORSSD.

carefully select several challenging scenarios in RSI-SOD. These scenarios are categorized and presented in Fig. 8. In the scenario of large salient regions (a), our method effectively highlights the entire salient regions, while other methods like ACCoNet and ERPNet tend to discard certain constituent salient regions or introduce background clutter. Our method demonstrates a strong scale-adaptive capability, enabling robust performance in scenarios involving both large salient regions and multiple small salient objects (b). For salient objects with irregular topologies (c), our method excels in revealing the overall structure and complete salient regions. This can be attributed to our method's exploitation of context information from a global-view perspective, particularly through cross-image contexts, which comprehensively enhances the representation. Notably, DAFNet, which fully leverages spatial global attention, also achieves good performance in this challenging scenario, underscoring the importance of global-view context in addressing such cases. The scenarios involving salient objects with low contrast (d) and salient objects with background interference (e) are common challenges encountered in RSIs. Our method successfully handles these scenarios, while representative RSI-SOD methods, such as ACCoNet and MCCNet, are more susceptible to background clutter. The results obtained in these challenging scenarios validate the effectiveness of the cross-image context exploitation employed in our method.

C. Ablation Study

1) Effectiveness of SAF, CDM, and ISM: We perform ablation experiments to measure the contributions of the proposed modules, namely SAF, CDM, and ISM, in MCP-Net. The results are presented in Table III. To establish a baseline model (referred to as "BL"), SAF, CDM, and ISM are removed from the final architecture of MCP-Net. Subsequently, the proposed modules are gradually integrated into the baseline model, leading to the second to fourth rows in Table III. The performance improves steadily with the integration of each key component. It is important to note that the baseline model achieves moderate scores and performs worse than most RSI-SOD methods. Therefore, the improvements observed due to the proposed modules are significant and substantial. Notably, integrating only CDM (third row) results in a considerable improvement over the baseline model. Specifically, on EORSSD, the integration of CDM improves S_m and MAE by 1.88% and 0.0036, respectively. On ORSSD, the improvements are even more pronounced, with an increase of 2.84% in S_m and 0.0069 in MAE. These remarkable improvements highlight the necessity of utilizing cross-image contexts in RSI-SOD and demonstrate the effectiveness of the designed memory mechanism in CDM.

Furthermore, more comprehensive ablation analyses are performed by exploring different combinations of the proposed

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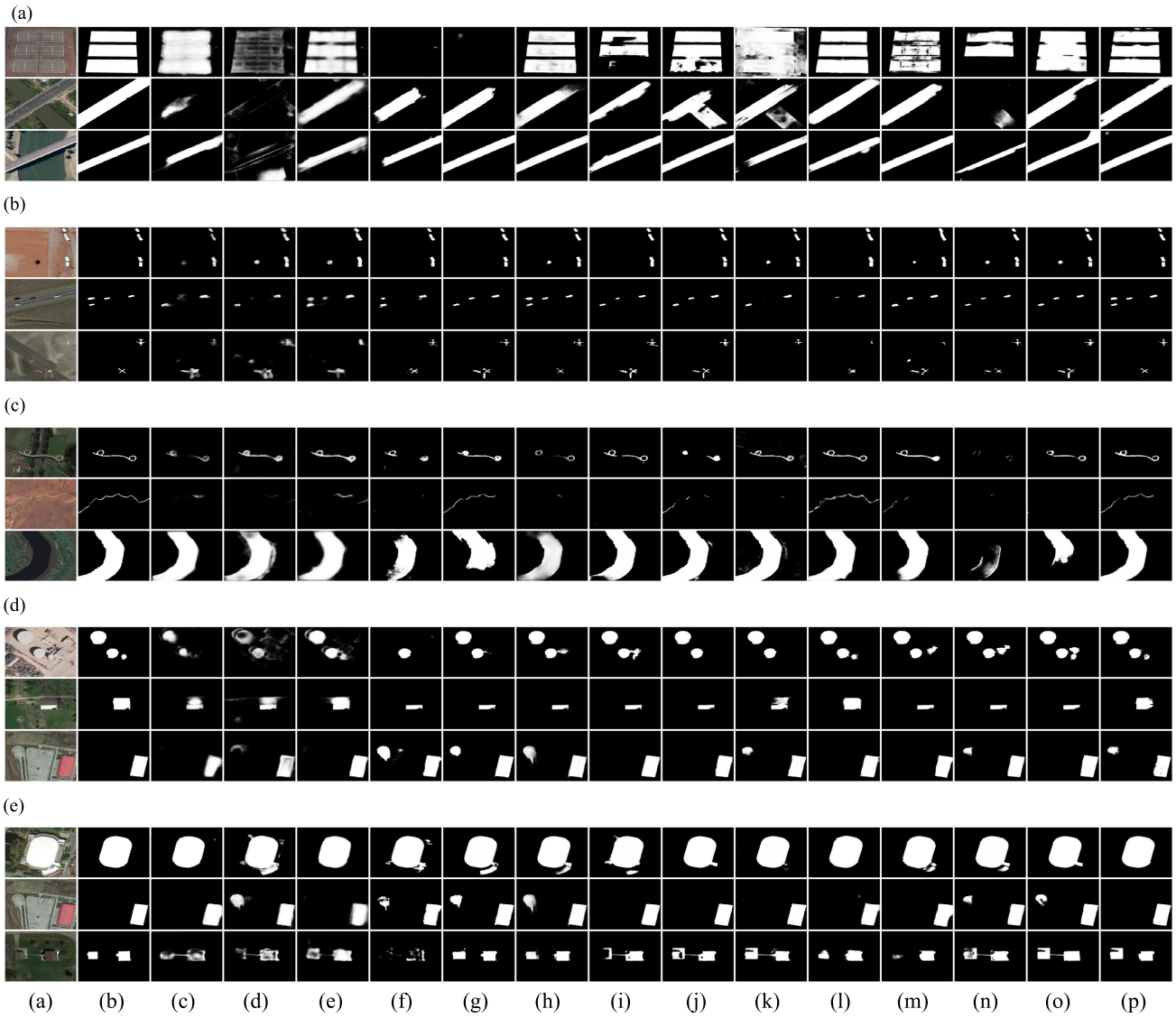


Fig. 8. Visual comparisons with 13 state-of-the-art methods, including four CNN-based NSI-SOD methods, and seven CNN-based RSI-SOD methods. (a) Optical RSIs. (b) GT. (c) DSS. (d) PoolNet. (e) EGNet. (f) LVNet. (g) EMFINet. (h) MJRBM. (i) FSMINet. (j) MCCNet. (k) SARNet. (l) DAFNet. (m) SeaNet. (n) ERPNet. (o) ACCoNet. (p) OURS.

TABLE III
ABLATION STUDY ON THE KEY COMPONENTS OF MCP-NET

BL	SAF	CDM	ISM	EORSSD		ORSSD	
				$S_m\uparrow$	$M\downarrow$	$S_m\uparrow$	$M\downarrow$
✓				.9127	.0116	.9040	.0208
✓	✓			.9267	.0086	.9375	.0111
✓		✓		.9299	.0080	.9297	.0139
✓			✓	.9282	.0080	.9299	.0133
✓	✓	✓		.9339	.0074	.9335	.0122
✓	✓		✓	.9290	.0086	.9396	.0103
✓		✓	✓	.9320	.0074	.9357	.0108
✓	✓	✓	✓	**.9373**	**.0070**	**.9433**	**.0090**

modules (fifth to eighth rows). In general, significant improvements are observed when comparing the combination variants with those that integrate a single module. For example, the variant that integrates both CDM and ISM outperforms the variant that integrates only CDM or ISM. Similar observations are made in the sixth and seventh rows. Finally, as depicted in the eighth row, the complete MCP-Net achieves the best scores among all the alternatives. It improves the baseline model by 2.70% in S_m and 0.0046 in MAE on EORSSD, as well as by 4.35% in S_m and 0.118 in MAE on ORSSD. These improvements further validate the effectiveness of the proposed modules in enhancing the performance of MCP-Net.

2) Ablation on SAF: The ablation study of the SAF module on ORSSD and EORSSD is presented in Table IV. Several variants are obtained by removing key components in SAF, namely: 1) shared weights in the spatial attention layer ("w/o. S-SA"); 2) low-level spatial attention branch ("w/o. L-SA"); 3) high-level spatial attention branch ("w/o. H-SA"); and 4) channel attention ("w/o. CA"). From the results, it can be inferred that weight-sharing contributes to performance improvement, thereby validating the effectiveness of the weight-sharing mechanism in spatial attention. Additionally, removing the low-level attention branch causes more significant performance degradation compared to removing

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TABLE IV
ABLATION STUDY ON SAF

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
SAF	**.9433**	.0090	**.9135**	**.9373**	**.0070**	**.8868**
SAF (w/o. S-SA)	.9374	.0099	.9081	.9324	.0074	.8809
SAF (w/o. L-SA)	.9403	.0099	.9109	.9335	.0076	.8826
SAF (w/o. H-SA)	.9422	**.0086**	.9105	.9353	.0076	.8843
SAF (w/o. CA)	.9405	.0099	.9103	.9335	.0078	.8824

TABLE V
ABLATION STUDY ON CDM

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
CDM (FM)	.9403	.0097	.9106	.9320	.0075	.8806
CDM (BM)	.9353	.0099	.9053	.9340	.0078	.8823
CDM (FM & BM)	**.9433**	**.0090**	**.9135**	**.9373**	**.0070**	**.8868**
DPH (No. Sup)	.9404	.0099	.9108	.9349	.0070	.8842
DPH (S. Sup)	.9418	.0092	.9099	.9356	.0074	.8851
DPH (O. Sup)	.9372	.0110	.9049	.9354	.0073	.8839
DPH (S/O. Sup)	**.9433**	**.0090**	**.9135**	**.9373**	**.0070**	**.8868**

the high-level attention branch. This observation highlights that the low-level attention mechanism is more effective in preserving informative details.

3) Ablation on CDM: The ablation study of the CDM module is presented in Table V. First, the efficacy of employing foreground and background memory banks to store cross-image contextual information is investigated. Three variants are examined: 1) foreground memory only ("FM"); 2) background memory only ("BM"); and 3) both foreground and background memory ("FM & BM"). The results indicate that employing both foreground and background memory is more effective than using a single memory, demonstrating the effectiveness of the dual memory modules in cross-image context modeling. Second, the supervision strategies in the DPH module are studied. Four different strategies are deployed: 1) no supervision ("No. Sup"); 2) supervision using scene annotations ("S. Sup"); 3) supervision using object annotations ("O. Sup"); and 4) supervision using both scene and object annotations ("S/O. Sup"). It is worth noting that using no supervision slightly degrades the performance compared to using both scene and object supervision. This suggests that memory queries, such as scene and foreground predictions, can be learned automatically and accurately without annotations during training.

Multiple memory units are adopted in CDM to capture scene-dependent content variations. Therefore, ablation experiments are conducted to determine the optimal number of memory units. The ablation results, presented in Fig. 9, show the evaluation of S_m, $F_\beta^{\max}$, and $E_\varepsilon^{\max}$ on EORSSD and ORSSD using the memory unit number 1, 3, 5, 7, and 9. It is observed that the metric scores increase as the number of memory units increases, but they tend to saturate at 5 and stop increasing. Consequently, the number of memory units is set to 5 in the final architecture.

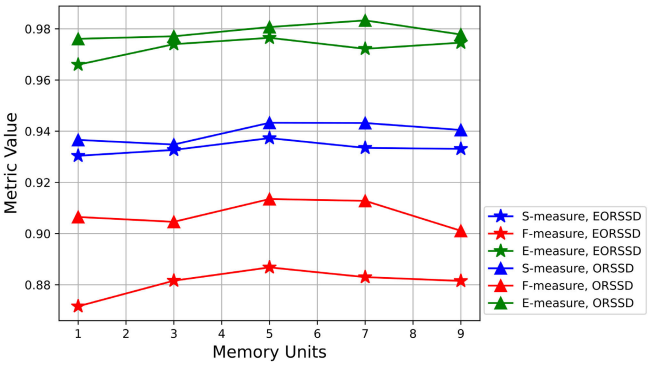


Fig. 9. Ablation experiments on the number of memory units.

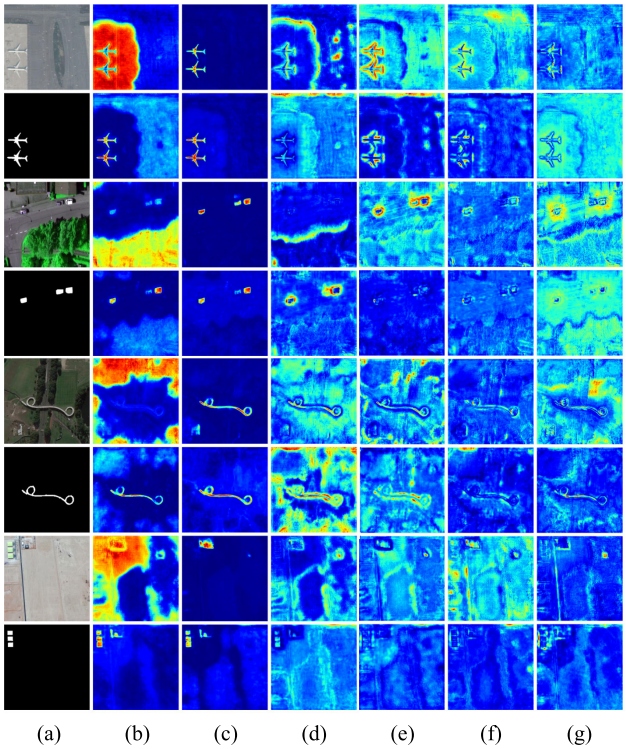


Fig. 10. Visualization of feature maps. (a) Optical RSIs and GTs. The odd rows show the optical RSIs, while the even rows show the corresponding GTs. From (b) to (g) PCA maps corresponding to the first to the sixth principal components in order. The odd rows show the visualization of our model without CDM, while the even rows show the visualization of our model with CDM.

Furthermore, to better understand the effectiveness of utilizing cross-image context, feature maps of the proposed method are visualized. Principal component analysis (PCA) is applied to the final aggregated feature in the first level, i.e., $\mathcal{O}_{ii}^1 + \mathcal{O}_{ci}^1$, for channel reduction. The first six principal components are retained for visualization. A comparison is conducted between feature representations learned with the CDM module and those learned without it. The visualization comparison, presented in Fig. 10, demonstrates that CDM can obtain more discriminative representations and suppress background clutter, thus verifying the effectiveness of exploiting memory-based cross-image contextual information in RSIs.

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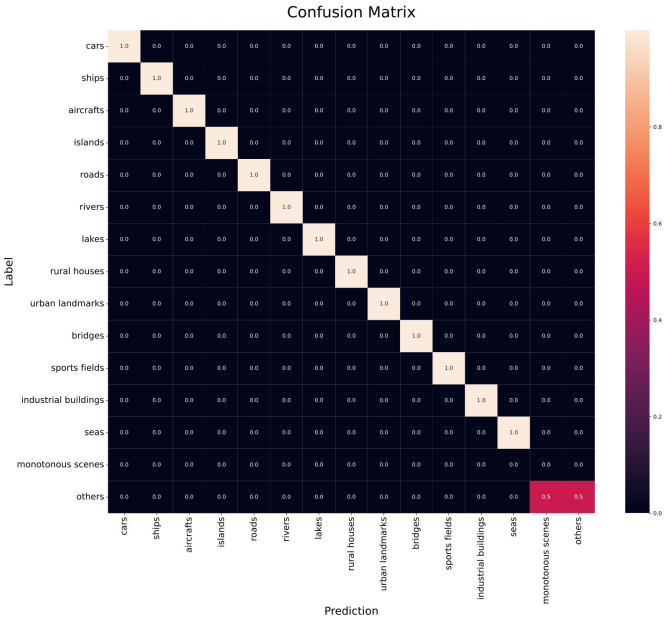


Fig. 11. Visualization of confusion matrix depicting scene-category predictions based on memory content within the CDM.

TABLE VI
ABLATION STUDY ON ISM

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
ISM	**.9433**	**.0090**	**.9135**	**.9373**	**.0070**	**.8868**
ISM (w/o. ASPP)	.9406	.0095	.9079	.9325	.0074	.8807
ISM (w/o. Gate)	.9375	.0095	.9066	.9354	.0074	.8822
ISM (w/o. Bidir)	.9396	.0101	.9097	.9336	.0080	.8808

We also construct a confusion matrix to evaluate the discriminative ability of the memory representations within the CDM, as illustrated in Fig. 11. Predictions are generated using a linear model trained on the foreground memory content to categorize the scene types. The results reveal that memory representations exhibit a strong ability to distinguish between scene categories, albeit with the exception that "monotonous scenes" tend to be confused with "others."

4) Ablation on ISM: The ablation study of the ISM module is presented in Table VI. The effectiveness of the ISM module is verified by removing two key components: the ASPP module ("w/o. ASPP") and the gate attention mechanism ("w/o. Gate"). It can be observed that both components significantly contribute to performance improvement. Specifically, removing the ASPP leads to a greater degradation on EORSSD, while removing the gate attention mechanism leads to a greater degradation on ORSSD. These findings highlight the importance of the ASPP module and the gate attention mechanism in improving performance. Furthermore, the bidirectional ISM connections are verified by removing the bottom-up path from the shallow layers to the deep layers ("w/o. Bidir"). The results demonstrate that the bottom-up path contributes to performance improvement, providing evidence for the efficacy of bidirectional memory propagation.

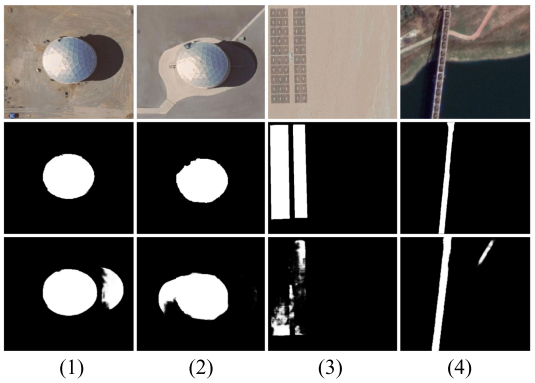


Fig. 12. Representative failure cases of the proposed MCP-Net. From top to bottom rows: optical RSIs, GT, results produced by our model.

D. Failure Case Analyses

We thoroughly investigate the failure cases of our proposed MCP-Net, as they provide valuable insights for potential future improvement. In Fig. 12, we present some representative failure cases of MCP-Net. These failure cases highlight specific challenges that our method encounters. In case (1) and (2), it can be observed that MCP-Net is susceptible to shadow interference and region connection. These scenarios pose difficulties in accurately detecting salient regions, potentially leading to redundancy incorporation and incorrect segmentation. Case (3) demonstrates that MCP-Net encounters challenges when dealing with low-contrast scenes where large salient regions need to be emphasized. Due to the lack of clear visual cues in such scenes, the proposed memory mechanism in our method may struggle to detect and highlight these regions effectively. Case (4) reveals a situation where MCP-Net tends to highlight extra objects (e.g., roads) while detecting the prominent object (e.g., a bridge) in this RSI. This particular case poses a significant challenge since roads and bridges are considered salient objects in different scenarios. It also exhibits the complexities of defining salient objects across various contexts and indicates the importance of constructing RSI-SOD datasets with clearer guidelines for salient objects in diverse scenarios.

V. CONCLUSION

In this article, we address the task of RSI-SOD by exploring the untapped potential of cross-image contextual information. To harness the power of cross-image contexts, we propose an MCP-Net that extracts dataset-level contextual information. Our MCP-Net incorporates a CDM designed to store and utilize dataset-level information. By leveraging foreground and background memory banks, CDM effectively captures intra-scene variations, improving salient object representations. Moreover, we enhance the representations by leveraging contextual information within individual images through an ISM propagation module. To preserve RSI details, we introduce an SAF module that aligns the cross-network-level features. Extensive evaluation on three public RSI-SOD benchmarks demonstrates the superior performance of our proposed method compared to state-of-the-art approaches. In the future, there is significant potential to improve the performance by

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combining more robust within-image representations with memory mechanisms. Furthermore, the proposed memory mechanisms can be integrated into more pixel-level prediction tasks, such as edge and skeleton detection, for effective cross-image information retrieval.

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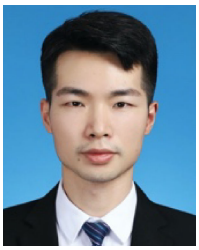
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